

Chengxiang Huang

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[Google Scholar](#)

EDUCATION

University of Toronto

➤ PhD in Electrical & Computer Engineering

Sept. 2026– Present

Beijing University of Posts & Telecom

➤ B.Eng. in Telecommunication Engineering with Management

Sept. 2021– June 2025

PUBLICATIONS & MANUSCRIPTS

- **Chengxiang Huang**, et al., “Concentrated Gradients Amplify Forgetting: Dominant-direction Projection for Continual Multimodal Learning,” *under review at Neural Information Processing Systems (NeurIPS)*, 2026.
- Yi Wang, Haopeng Zhang, **Chengxiang Huang**, et al., “LoopArena: Benchmarking Models as Runtime Controllers for Loop Engineering,” *arXiv preprint arXiv:2608.28281*, 2026. [\[arXiv\]](#) [\[code\]](#) [\[project\]](#)
- Lin Huang, **Chengxiang Huang**, et al., “E2Former-V2: On-the-Fly Equivariant Attention with Linear Activation Memory,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2026. [\[arXiv\]](#) [\[code\]](#)
- Shuangjie Fan, **Chengxiang Huang**, et al., “Beyond Adapter Expansion: Packing Multiple Tasks into Spare LoRA Capacity,” in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
- **Chengxiang Huang**, Yake Wei, Zequn Yang and Di Hu, “Adaptive Unimodal Regulation for Balanced Multimodal Information Acquisition,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [\[arXiv\]](#) [\[code\]](#)
- **Chengxiang Huang**, Bingyan Liu, “PA³Fed: Period-Aware Adaptive Aggregation for Improved Federated Learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

RESEARCH EXPERIENCE

Multimodal Continual Learning via Dominant-direction Projection

Feb. 2026 - Sept. 2026

Alibaba Group | Beijing

- Conducted research on LoRA-based continual instruction tuning of MLLMs, analyzing how sequential audio/image/video tasks overwrite previously learned knowledge.
- Proposed **DiGPro** (Dominant-direction Gradient Projection), which extracts an adaptive subspace from a short gradient buffer and attenuates the parallel component at each optimizer step, without storing past-task data or subspaces. First-authored a paper currently under review at NeurIPS 2026.

Transformer-based Equivalent Molecular Foundation Model Development

Sept. 2025 - Feb. 2026

Ubiquant | Beijing

- Participated in optimizing a large-scale **Transformer-based molecular foundation model**, focusing on efficient architecture design and distributed training stability.
- Accelerated model computation using **Triton kernels** and optimized memory usage for high-throughput multi-GPU training.

Multimodal Imbalance Learning from Information Perspective

April 2024 - Dec. 2024

Renmin University of China | Beijing

- Conducting in-depth research in **multimodal learning** focusing on how different modalities acquire and interact with information during training.
- Co-authored a paper, *Adaptive Unimodal Regulation for Balanced Multimodal Information Acquisition (InfoReg)*, which introduces a **Fisher Information–based regulation mechanism** to balance modality-specific information acquisition.